NIC KANE

Data Science & Machine Learning Graduate Student
San Francisco, CA | nicmuellerkane@gmail.com
nickane12.github.io/web | github.com/NicKane12 | linkedin.com/in/nic-kane

PROFESSIONAL SUMMARY

M.S. Statistical Data Science candidate with hands-on experience in machine learning, statistical modeling, audio classification, and time-series forecasting using Python and R. Nearly nine years of teaching experience supports clear technical communication.

TECHNICAL SKILLS

Languages & Tools: Python (pandas, NumPy, librosa, PyTorch, scikit-learn), R, SQL, Excel, Tableau, Power BI
Methods: Classification, Regression, Random Forests, SVMs, Boosting, Neural Networks, Vision Transformers, Self-Attention, MFCCs, Audio Feature Engineering, Time Series (ARIMA, SARIMA, ETS), Cross-Validation, ROC/AUC, Macro F1

APPLIED DATA SCIENCE PROJECTS

Music Genre Classification from Audio Features

- Built an end-to-end audio classification pipeline in Python using the Free Music Archive dataset, extracting progressively richer feature sets from 30-second tracks.
- Engineered and compared time-domain, spectral, MFCC, beat, and frequency-band features; evaluated logistic regression, random forest, SVM, gradient boosting, and neural-network classifiers using accuracy and macro F1.

Predicting IMDb Movie Ratings from Metadata

- Built and compared linear regression, ridge/lasso, support vector regression, and random forest models in Python to predict IMDb ratings from runtime, release year, and genre across 100K+ films.
- Engineered features through missingness analysis, multi-label genre encoding, log-transformed vote counts, and runtime outlier filtering; tuned models with GridSearchCV and cross-validation.
- Achieved the best performance with a tuned random forest (RMSE = 1.20), improving over baseline OLS while preserving interpretable variable-importance rankings.

Time Series Analysis of U.S. Vinyl Record Sales

- Analyzed longitudinal U.S. vinyl sales data for trend and stationarity using ACF/PACF; built and compared ARIMA and exponential-smoothing models using AIC and residual analysis.
- Generated short- and long-term forecasts with confidence intervals and interpreted forecast uncertainty for demand-forecasting applications.

PROFESSIONAL EXPERIENCE

Machine Learning Intern | Synth Rails — Sep 2026 – Present

- Developed foundational understanding of vision transformer architecture, including patch embeddings, self-attention, and transformer-based vision workflows.
- Studied flow matching and its role in generative modeling through technical reading, guided discussion, and research preparation.

Graduate Teaching Assistant, Prelude to Calculus, **Calc I Lab** | San Francisco State University — Aug 2025 – Dec 2026

- Taught a preparatory mathematics course for STEM majors building readiness for calculus-level coursework using small-group collaborative learning.
- Designed and administered assessments and a cumulative project to evaluate student readiness.
- Translated complex quantitative concepts into clear, structured explanations.

Instructor, Mathematics & Music | Fusion Academy, San Francisco Bay Area — Feb 2019 – Aug 2026

- Delivered individualized mathematics instruction from 6th grade through calculus, adapting explanations to diverse learning needs.
- Communicated technical concepts clearly to students, families, and administrators while maintaining detailed progress documentation.

Additional: Math & Physics Instructor, Lydian Academy; Senior Leader & Math Workshop Facilitator, Cal Poly San Luis Obispo.

EDUCATION

M.S. in Statistical Data Science — Expected December 2026
San Francisco State University, San Francisco, CA

B.S. in Electrical Engineering, Minor in Mathematics — 2018
Cal Poly, San Luis Obispo, CA